

Ex 3: consider $\boxed{\min f(x) \text{ s.t. } \underline{g_i(x) \leq 0}, i=1, \dots, 2}$

$$L(x, \lambda) = f(x) + \lambda_1 g_1(x) + \lambda_2 g_2(x)$$

$\lambda_i \geq 0$

$$d(\lambda) = \min_x L(x, \lambda)$$

Assumption $f(x), g_1(x), g_2(x)$ convex function

Start from $(x^{(0)}, \lambda^{(0)})$
compute $\underline{x^{(1)}}$ = argmin $L(x, \lambda^{(0)})$

Then update λ .

Dual subgradient
Ascent Algorithm

$$\left. \begin{aligned} \lambda_1^{(1)} &= \text{proj}_{\geq 0} [\lambda_1^{(0)} + \alpha g_1(x^{(1)})] \\ \lambda_2^{(1)} &= \text{proj}_{\geq 0} [\lambda_2^{(0)} + \alpha g_2(x^{(1)})] \end{aligned} \right\}$$

For maximizing $d(\lambda)$, the above is dual ascent algorithm.

$d(\lambda)$ dual function (concave).

- $d(\lambda)$ is convex function. we can talk about subgradient.

claim: let $\underline{h(\lambda)} := -d(\lambda)$
 $= -f(x^*(\lambda)) - \lambda_1 g_1(x^*(\lambda)) - \lambda_2 g_2(x^*(\lambda))$
 where $x^*(\lambda) = \arg\max_x L(x, \lambda)$. Then we have:

$$s^*(\lambda) := \begin{bmatrix} -g_1(x^*(\lambda)) \\ -g_2(x^*(\lambda)) \end{bmatrix} \in \partial h(\lambda).$$

Basically, we need to show that:

$$h(\underline{\sigma}) \geq h(\lambda) + s^*(\lambda)(\underline{\sigma} - \lambda), \forall \sigma, \lambda \geq 0$$

proof:

$$\begin{aligned} h(\lambda) &= -f(x^*(\lambda)) - \lambda_1 g_1(x^*(\lambda)) - \lambda_2 g_2(x^*(\lambda)) \\ h(\sigma) &= -f(x^*(\sigma)) - \sigma_1 g_1(x^*(\sigma)) - \sigma_2 g_2(x^*(\sigma)) \end{aligned}$$

not $x^*(\sigma)$ maximizes $= \max_x -L(x, \sigma)$

$L(x, \sigma); \leftarrow x^*(\lambda) \geq -L(x^*(\lambda), \sigma)$

does not.

$$= -f(x^*(\lambda)) - \sigma_1 g_1(x^*(\lambda)) - \sigma_2 g_2(x^*(\lambda))$$

$$\begin{aligned} h(\sigma) &\geq h(\lambda) - (\sigma_1 - \lambda_1) g_1(x^*(\lambda)) \\ &\quad - (\sigma_2 - \lambda_2) g_2(x^*(\lambda)) \end{aligned}$$

$$= h(\lambda) + \begin{bmatrix} -g_1(x^*(\lambda)) \\ -g_2(x^*(\lambda)) \end{bmatrix}^T \begin{bmatrix} \sigma_1 - \lambda_1 \\ \sigma_2 - \lambda_2 \end{bmatrix}$$

$$= h(\lambda) + s^*(\lambda)^T (\sigma - \lambda)$$